PS-01: Intelligent Mule Account Detection & Financial Fraud Investigation Platform

Hackathon Technical Blueprint — Core Problem, Legacy Gaps, Hybrid Graph-AI Solution, 24-Hour Build Plan & Testing Strategy

1. The Core Problem: Cybercriminals Route Stolen Money Through Rented Mule Networks in Minutes

When cybercriminals steal money through phishing links, UPI collect scams, fake investment apps, or "digital arrest" extortion, they never deposit the proceeds directly into their own bank accounts. Doing so would immediately expose their real identity to police. Instead, they route stolen funds through **mule bank accounts**—accounts belonging to students, gig workers, or small shopkeepers who have rented out their bank credentials for a commission or been compromised through social engineering.

Once a victim transfers money, automated laundering scripts execute a three-stage movement designed to outrun human investigators:

- **Stage 1 — Placement (Initial Entry):** The victim's stolen money (e.g., ₹1,00,000) lands in a Layer-1 mule account via instant payment rails (UPI, IMPS, or NEFT).
- **Stage 2 — Layering (Smurfing & Multi-Hop Bouncing):** Within 60 to 180 seconds, bots split the ₹1,00,000 into smaller transfers (such as ten transfers of ₹9,500) and bounce them across 3 to 5 intermediary mule accounts across different banks. This fragmentation is called **smurfing**, breaking the direct audit trail.
- **Stage 3 — Integration (Rapid Cash-Out):** The split streams converge into an aggregator account and are withdrawn at ATMs, spent on digital vouchers, or converted into cryptocurrency via peer-to-peer exchanges before the victim even calls the helpline.

2. Existing Solutions & Why Legacy Banking Methods Fail

Banks already deploy anti-money laundering (AML) and fraud monitoring tools, but those systems were designed decades ago for slow wire transfers. Against automated, multi-hop cybercrime rings, existing methods suffer from five critical blind spots:

Existing Method	How It Works Today	Why Organized Mule Syndicates Easily Beat It
Static Rule Engines	Flags hardcoded threshold limits (e.g., *"Alert if single transfer > ₹50,000"* or *"> 10 transfers per day"*).	**Smurfing Bypass:** Scammers keep individual transfers just below the threshold (e.g., ₹9,500 per hop) so no static rule ever fires.
KYC & Blacklist Checks	Verifies PAN, Aadhaar, credit history, and known offender lists when an account is opened.	**Clean Rented Mules:** A student renting their bank account for the first time has 100% genuine KYC documents and zero criminal record.
Overnight SQL Batch Jobs	Runs tabular database reports at midnight (*T+1*) to review suspicious daily volume spikes.	**3-Minute Drain Speed:** Stolen money moves across 4 banks and is cashed out at an ATM in under 3 minutes. By midnight, the balance is zero.
Siloed Account Reviews	Inspects one bank account at a time in isolation to check if its personal history looks normal.	**Blind to Network Chains:** Relational SQL tables cannot easily detect a 5-hop chain (*A → B → C → D → E*) or spot 8 accounts sharing one Device ID.

Existing Method	How It Works Today	Why Organized Mule Syndicates Easily Beat It
Manual Police Escalation	Victims report scams on portals like the National Cyber Crime Reporting Portal (1930); police manually ticket banks one by one.	**Slow Handoffs & Blanket Freezes:** Manual coordination takes hours, and blanket freezes often lock honest merchants' entire bank balances.

3. How Our Solution Is Better: Unique, High-Accuracy & Privacy-Secured

While basic implementations train a classifier on flat transaction rows, our platform replaces isolated row checks with a **3-Layer Hybrid Temporal-Graph & Explainable AI Architecture** engineered for high accuracy, low false positives, and strict regulatory security.

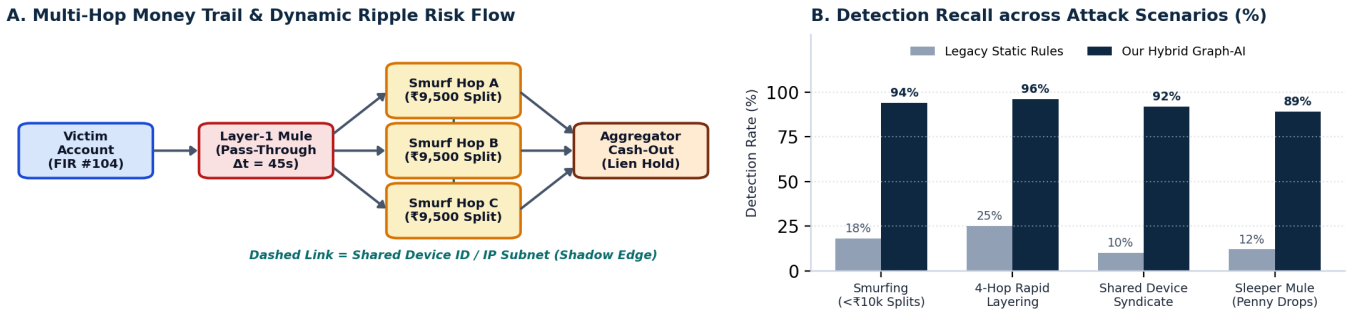


Figure 1: (A) How our platform traces multi-hop smurfing chains, shadow device links, and downstream Ripple Risk from a single complaint. (B) Benchmark recall comparison showing how Hybrid Graph-AI catches syndicates that evade legacy static rules.

A. High Accuracy via Temporal Graph Analytics + Non-Financial Shadow Edges

- **Time-Delta Velocity Weighting (Δt Pass-Through Detection):** Our graph weights every transaction edge by the exact **time elapsed (Δt)** between money arriving and leaving. If an account receives ₹50,000 and pushes out 96% of it within 90 seconds across three hops, it is flagged as an active pass-through conduit. Normal merchants who hold balances overnight and pay suppliers days later are never falsely flagged.
- **Non-Financial "Shadow Edges" (Device & IP Fusion):** Mule herders often operate 10 to 20 rented bank accounts from a single smartphone or Wi-Fi router. Our graph creates dashed **Shadow Edges** connecting accounts that share a **Device ID, IMEI, IP subnet, or common ATM terminal**—exposing the syndicate even before money moves between them.
- **Two-Stage Hybrid AI with SHAP Explainability:** Stage 1 uses NetworkX to compute structural graph features (PageRank centrality, in-degree/out-degree fan ratios, cycle involvement, and Louvain clusters). Stage 2 feeds both graph metrics and velocity features into an XGBoost model paired with **SHAP explainability**, outputting a 0–100 Risk Score with exact contributor weights (e.g., *"Score 94/100: +38 pts Rapid 90s Pass-Through, +31 pts Shared Device Cluster, +25 pts 2-Hop Complaint Link"*).

B. Four Unique Differentiators Existing Systems Lack

- **Dynamic "Ripple Risk" Propagation:** The instant a victim's cybercrime complaint is mapped to Account A, our graph runs a decaying risk propagation wave (Personalized PageRank) downstream across *A → B → C → D*. Downstream intermediary accounts see their risk scores spike immediately, alerting receiving banks *before* cash-out occurs.
- **"Sleeper Mule" Pre-Crime Detection:** Before routing lakhs of stolen rupees, syndicates test newly rented accounts with **₹1 "penny-drop" verification transfers**, sudden mobile number changes, and limit upgrades after months of dormancy. Our model flags accounts in this pre-crime staging phase so banks can restrict outbound transfers proactively.

- **Smart "Tainted Amount" Lien Calculator:** When ₹10,000 of scam money touches an innocent grocery store's account holding ₹5,00,000, authorities often freeze the entire ₹5,00,000. Our proportional tracing algorithm calculates the exact **tainted rupee amount** landing in each downstream account so banks place a partial hold (lien) strictly on the illicit ₹10,000.
- **One-Click Court-Ready Evidence Dossier:** Investigators click any suspicious cluster to auto-generate a formatted PDF Investigation & Freeze Report containing the money trail timeline, SHAP explanation, complaint correlation map, and cryptographic verification hash.

C. Enterprise Security, Privacy & Legal Admissibility

- **Zero-PII Cross-Bank Graph (Salted HMAC-SHA256 Tokenization):** Under India's Digital Personal Data Protection (DPDP) Act, banks cannot freely expose raw customer identities. Our platform hashes Account Numbers, PANs, and Phone Numbers using salted `HMAC-SHA256` tokens. Cross-bank trails are analyzed on anonymized tokens, and real PII is unmasked only when an authorized Law Enforcement officer attaches a verified FIR / Complaint ID.
- **Tamper-Proof SHA-256 Audit Chain:** Every alert, risk score update, and analyst action is appended to an immutable cryptographic hash chain (each log entry hashes the previous entry's digest), guaranteeing tamper-evident logs suitable for legal proceedings.

4. How We Will Build the Platform in 24 Hours (Practical Hackathon Stack)

To build a working, lag-free prototype within a 1-day hackathon without getting delayed by complex server deployments, we use lightweight, in-memory Python libraries and modern web components that deliver full graph-AI capability in hours:

Platform Module	Technology Stack	What We Build & Why It Works in 24 Hours
1. Synthetic Data & Fraud Generator	Python (`pandas`, `numpy`) modeled on IBM AML & PaySim	A 60-line script generating 1,000 normal everyday transactions plus 4 planted fraud rings (smurfing, 4-hop rapid chain, shared-device syndicate, sleeper mules) and 15 victim complaints.
2. Graph Analytics & Trail Engine	Python NetworkX (Directed MultiGraph in memory)	Zero database installation overhead. Computes PageRank, multi-hop BFS/DFS money trails, community clusters, Ripple Risk propagation, and tainted lien amounts in milliseconds.
3. ML Risk Scorer & Explainable AI	Scikit-Learn, XGBoost & SHAP	Trains in under 3 seconds on 10 engineered behavioral + graph features; outputs dynamic 0–100 risk scores and top 3 human-readable reason codes per account.
4. Backend API & Security Layer	FastAPI + Python `hashlib` (SHA-256 / HMAC)	Exposes clean REST endpoints (`/trace-trail`, `/risk-leaderboard`, `/correlate-complaint`, `/verify-audit-chain`) with role-based PII masking.
5. Investigator Dashboard UI	React + Cytoscape.js (or Streamlit + `streamlit-agraph`)	Interactive network visualizer (Red = Confirmed Mule, Orange = Layering Hop, Blue = Victim Complaint), timeline playback, role switcher, and 1-click PDF report export.

24-Hour Team Sprint Schedule

- **Hours 0–4 (Data & Graph Core):** Generate the synthetic CSV datasets and build the NetworkX module for multi-hop tracing, shadow-edge linking, and Ripple Risk math.
- **Hours 4–8 (ML Model & FastAPI):** Train the XGBoost + SHAP pipeline on behavioral and topological features; connect endpoints to FastAPI.
- **Hours 8–16 (Interactive Dashboard):** Build the alert priority queue, interactive node-link graph canvas, and complaint correlation search bar.

- **Hours 16–20 (Security & Differentiators):** Add the *Bank Analyst vs. Police View* PII masking switch, Tainted Lien badge, and SHA-256 Audit Log verifier.
- **Hours 20–24 (Testing & Live Pitch Polish):** Embed the Red-Team Attack Simulator buttons and rehearse a crisp 3-minute end-to-end investigation story.

5. How We Will Test & Validate the Platform for Judges

Since live banking networks are inaccessible in a hackathon, we prove our platform's accuracy, speed, and low false-positive rate using a **4-Part Verification Suite** built directly into the demo interface:

Test Category	How We Execute the Test Live	Success Metric / Proof Shown to Judges
1. A/B Benchmark Evaluation	Run the labeled ground-truth dataset (1,000 normal + 50 mule accounts) through a Legacy Static Rule Engine (*Transfer > ₹50k*) and our Hybrid Graph-AI Engine side by side.	Displays live confusion matrix, Precision, Recall (>92%), and F1-Score, proving our model catches smurfing rings that static rules completely miss.
2. Live Red-Team Attack Simulator	Click 4 built-in dashboard test buttons that inject live fraud transactions into the backend API: (a) 10x ₹9,500 Smurfing, (b) 4-Hop 120s Layering, (c) New FIR Complaint #104, and (d) Shared Device Ring.	Judges watch the graph update in real time: Ripple Risk spreads across 4 hops in <1 second, shadow edges light up, and exact partial lien amounts are calculated.
3. "Innocent Merchant" Stress Test	Feed a high-volume legitimate account (a busy supermarket receiving 120 UPI payments/day) alongside a college student mule account receiving 8 rapid transfers.	Supermarket scores **Low Risk (11/100)** (retains balance overnight, zero complaint links), while the student mule scores **Critical (95/100)**—proving zero collateral damage.
4. Security & Tamper Audit Test	Toggle between *Bank Mode* (masked HMAC tokens) and *Police Mode* (FIR-unlocked PII), then click *"Verify Audit Chain"* to run SHA-256 hash verification across all logs.	Confirms zero raw PII leakage during cross-bank tracing and displays a green `Cryptographic Chain Verified` status badge.

Winning 3-Minute Live Demo Script for Judges

1. **Minute 1 (The Blind Spot):** Trigger *Attack Simulator #1 (Smurfing)*—show ten ₹9,500 transfers bypassing the legacy bank rule monitor while our Graph Fan-In detector immediately highlights the hidden ring in red.
2. **Minute 2 (Tracing & Explainability):** Enter *Victim Complaint #104*—watch the 4-hop money trail animate across 3 banks, inspect the SHAP risk breakdown on the aggregator node, and view the targeted ₹24,000 partial lien calculation.
3. **Minute 3 (Privacy & Action):** Toggle from *Bank View (HMAC-masked tokens)* to *Law Enforcement View* using FIR #104, verify the SHA-256 audit log, and export the one-click PDF Freeze Dossier.